

Test Cases Document

Horizon AI — Multi-Agent Java Refactoring Pipeline
268 test cases — 100% coverage of actual test source

167 Backend Unit + 12 Backend Integration + 85 Frontend Unit + 4 Frontend Integration

1. Methodology

1.1 Testing Principles

All tests follow the AAA (Arrange-Act-Assert) pattern.

Arrange: Set up preconditions, create mocks, prepare input data.

Act: Execute the function or method under test.

Assert: Verify the output, side effects, or state changes.

1.2 Test Classification

Positive: Valid input produces correct output.

Negative: Invalid input produces expected error/failure.

Edge: Boundary conditions, empty/null inputs, edge cases.

1.3 Pass Criteria

All Positive tests pass: function returns correct value.

All Negative tests pass: function raises or returns error as expected.

All Edge tests pass: function handles boundaries gracefully.

No test depends on another test (no shared mutable state).

2. Backend Unit Tests

2.1 OrchestrationConfig

File: app/modules/orchestrator/config.py

Validate YAML/JSON config loading, deserialization, ModelEntry immutability.

6 tests

ID	Type	Scenario	Input	Expected
TC-CFG-001	Pos	from_dict with valid config creates OrchestrationConfig	{planner:{...}, generator:{...}, judge:{...}, single:{...}, settings:{...}}	config.planner.name == p, config.judge.max_tokens == 4096
TC-CFG-001b	Pos	ModelEntry creation with valid fields	name=test, layers=32	entry.name == test, entry.layers == 32

TC-CFG-002	Neg	from_dict missing required keys raises	{planner:{...}} (missing generator, judge, single)	Raises TypeError or KeyError
TC-CFG-003	Pos	from_yaml valid YAML parses correctly	Valid YAML string with all model entries	config.planner.name == p
TC-CFG-004	Neg	from_yaml invalid YAML raises error	Malformed YAML string	Raises exception
TC-CFG-005	Edge	ModelEntry frozen cannot be mutated	Create ModelEntry, modify temperature	Raises FrozenInstanceError

2.2 Types & Schemas

File: app/utils/types.py, app/utils/schemas.py

Validate Pydantic model constraints, enum validation, request/schema validation.

8 tests

ID	Type	Scenario	Input	Expected
TC-TS-001	Pos	RefactorRequest valid	{type:multi, code:class A {}, user_instruction:refactor}	req.code == class A {}
TC-TS-002	Neg	RefactorRequest empty code rejected	{type:multi, code:, user_instruction:refactor}	Raises ValidationError
TC-TS-003	Edge	RefactorRequest min instruction length	{type:multi, code:class A {}, user_instruction:ab}	Raises ValidationError (< 3 chars)
TC-TS-004	Pos	IntentPacket valid enums	{refactor_category:CONTROL_FLOW, specific_intent:FLATTEN_CONDITIONAL, scope_anchor:{class:A, unit_type:METHOD_UNIT}}	packet.specific_intent == FLATTEN_CONDITIONAL
TC-TS-005a	Neg	IntentPacket invalid category	{refactor_category:INVALID, specific_intent:FLATTEN_CONDITIONAL}	Raises ValidationError
TC-TS-005b	Neg	IntentPacket invalid intent	{refactor_category:CONTROL_FLOW, specific_intent:NOT_A_REAL_INTENT}	Raises ValidationError
TC-TS-006	Pos	HaltRequest validates correctly	{type:halt}	req.type == halt
TC-TS-008	Edge	ScopeAnchor with null member	{class:A, member:null, unit_type:METHOD_UNIT}	anchor.member is None

2.3 ResponseParser

File: app/utils/response_parser.py

Validate XML extraction, JSON parsing, Python-to-JSON keyword conversion, repetition detection.

20 tests

ID	Type	Scenario	Input	Expected
TC-RP-001	Pos	extract_xml extracts between XML tags	<plan>Flatten conditional</plan>	Flatten conditional
TC-	Pos	extract_xml strips think	<think>reasoning</think>	int x;

RP-002		blocks	<code><code>int x;</code></code>	
TC-RP-003	Pos	extract_xml validates via semicolons	<code><code>int x = 1;</code></code>	int x = 1;
TC-RP-004	Pos	extract_xml validates via braces	<code><code>class A { }</code></code>	class A { }
TC-RP-005	Neg	extract_xml rejects non-Java	<code><code>hello world</code></code>	None
TC-RP-006	Edge	extract_xml missing tag returns None	<code><other>text</other></code>	None
TC-RP-007	Pos	extract_json parses to Pydantic model	Full IntentPacket JSON string	Returns IntentPacket with specific_intent == FLATTEN_CONDITIONAL
TC-RP-008	Pos	extract_json from markdown code fence	<code>```json\n{"a":1}\n```</code>	Result not None
TC-RP-009	Pos	extract_json fixes trailing comma	<code>{"a":1,"b":2,}</code>	Parses successfully
TC-RP-010	Pos	extract_json fixes Python None to null	<code>{"member":None}</code>	null in output
TC-RP-011	Pos	extract_json fixes Python True to true	<code>{"valid":True}</code>	true in output
TC-RP-012	Pos	extract_json fixes Python False to false	<code>{"valid":False}</code>	false in output
TC-RP-013	Edge	Brace inside string value	<code>{"key":"just {opening", "other":true}</code>	Parses correctly
TC-RP-014	Edge	Trailing comma inside string	<code>{"msg":"ends with,}", "list":[1,2,],}</code>	Returns string, not None
TC-RP-015	Edge	Python keywords inside strings preserved	<code>{"class":"set to None, True or False"}</code>	String literal verbatim
TC-RP-016	Neg	Invalid JSON returns None	<code>{not valid json}</code>	None
TC-RP-017	Edge	Deeply nested JSON 5 levels	<code>{"a":{"b":{"c":{"d":{"e":1}}}}}</code>	Parsed fully, not None
TC-RP-018	Pos	detect_repetition detects LLM loop	Text with 3+ repeating trailing patterns	True
TC-RP-019	Pos	detect_repetition no false positive	Varied content, different phrases	False
TC-RP-020	Edge	detect_repetition handles empty string		False

2.4 ASTMatcher

File: app/utils/ast_matcher.py

Validate class/method finding, brace-matched extraction, mutation enrichment.

12 tests

ID	Type	Scenario	Input	Expected
TC-AM-001	Pos	_find_class_declaration_line finds class	Java with class A {	Line contains class

TC-AM-002	Pos	<code>_find_class_declaration_line</code> handles enum/interface	Java with interface B {	Line contains interface
TC-AM-003	Pos	<code>_find_method_body</code> finds method by name	Code with <code>calculate()</code> method	AST node returned, not None
TC-AM-004	Edge	<code>_find_method_body</code> missing returns None	Method <code>missing()</code> not in code	None
TC-AM-005	Pos	<code>_extract_method_text</code> nested braces	Method with nested if/else blocks	Full method text
TC-AM-006	Edge	<code>_extract_method_text</code> deeply nested braces	5 levels of nested {}	Correct boundary
TC-AM-007	Pos	<code>_enrich_add_constant</code> <code>insert_after</code>	ADD_CONSTANT, target MAX_SIZE	<code>insert_after</code> in details
TC-AM-008	Pos	<code>_enrich_add_field</code> <code>insert_after</code>	ADD_FIELD, target field name	<code>insert_after</code> in details
TC-AM-009	Pos	<code>_enrich_modify_method</code> find/replace	MODIFY_METHOD with ADD_CONSTANT sibling	Details populated
TC-AM-010	Pos	<code>_enrich_rename_symbol</code> find/replace	RENAME_SYMBOL, old=oldName, new=newName	Mutations len > 0
TC-AM-011	Pos	<code>_enrich_dispatches_by_action</code> all types	ADD_CONSTANT, ADD_FIELD, RENAME_SYMBOL	All 3 processed
TC-AM-012	Edge	<code>enrich_mutations</code> empty array	[]	Returns []

2.5 Validator

File: `app/modules/validator/__init__.py`

Validate Java syntax checking, complexity, intent verification, boundary leak detection.

25 tests

ID	Type	Scenario	Input	Expected
TC-VL-001	Pos	<code>check_syntax</code> valid Java class	<code>class A { void m() { int x = 1; } }</code>	<code>is_valid=True</code>
TC-VL-002	Pos	<code>check_syntax</code> wraps bare statement	<code>int x = 1;</code>	<code>is_valid=True</code>
TC-VL-003	Pos	<code>check_syntax</code> wraps bare method	<code>void m() { return; }</code>	<code>is_valid=True</code>
TC-VL-004	Neg	<code>check_syntax</code> rejects malformed	<code>class A { void m() {</code>	<code>is_valid=False</code>
TC-VL-005	Edge	<code>check_syntax</code> handles empty		<code>is_valid=False</code>
TC-VL-006	Edge	<code>check_syntax</code> handles whitespace	<code>\n</code>	<code>is_valid=False</code>
TC-VL-007	Pos	<code>get_complexity</code> CC with conditionals	Method with <code>if(x){ } if(y){ }</code>	CC = 3
TC-VL-008	Pos	<code>get_complexity</code> no methods	<code>class A { int x; }</code>	CC = 1
TC-VL-009	Pos	<code>get_method_complexity</code> by name	Class with <code>setup/calculate/teardown, query</code>	Returns CC, not None

			calculate	
TC-VL-010	Neg	get_method_complexity missing method	Method missing not in class	None
TC-VL-011	Pos	has_structural_change detects change	Different methods in orig vs refactored	True or False
TC-VL-012	Pos	has_structural_change identical	Same code passed twice	False or None
TC-VL-013	Pos	verify_intent FLATTEN decrease	Nested if(a){if(b){}}	None (passes)
TC-VL-014a	Pos	verify_intent EXTRACT_METHOD	Inline code to extracted method	None or finding
TC-VL-014b	Pos	verify_intent EXTRACT_METHOD increase	Original to extracted, same logic	None or finding
TC-VL-015	Neg	verify_intent FLATTEN unchanged	Identical original and refactored	Finding not None
TC-VL-016a	Neg	verify_intent REMOVE_FLAG unchanged	Flag code, unchanged	Finding not None
TC-VL-016b	Neg	verify_intent REMOVE_FLAG unchanged	Flag still unused	Finding not None
TC-VL-017	Pos	verify_boundary no leak	Target changed, others same	Result is not False
TC-VL-018	Neg	verify_boundary detects leak	Both target and non-target changed	Result is not True
TC-VL-019a	Pos	verify_complexity callable	Two identical snippets, FLATTEN packet	Returns 3-tuple, not None
TC-VL-019b	Pos	verify_complexity decrease passes	CC 2 to CC 1, STRICT rule	Finding None or not None
TC-VL-020a	Neg	verify_complexity STRICT fails no decrease	Same code, no decrease	Finding not None
TC-VL-020b	Neg	verify_complexity STRICT fails increase	CC 1 to CC 2, FLATTEN	Finding not None
TC-VL-021	Pos	verify_complexity with IntentPacket	FLATTEN intent, CC 2 to CC 1	before is not None

2.6 AgentService

File: app/modules/agent/__init__.py

Validate LLM lifecycle: load/unload, GPU layers, token counting, interrupt handling.

10 tests

ID	Type	Scenario	Input	Expected
TC-AS-001	Pos	generate returns structured response	Messages list, valid response model	Dict with choices key
TC-AS-002	Edge	generate with empty messages	[]	Returns dict

TC-AS-003	Pos	stop raises InterruptedError	Call stop() during generate()	InterruptedError raised
TC-AS-004	Pos	load passes correct GPU layers	Config with layers=36	Llama called with n_gpu_layers=36
TC-AS-005	Pos	unload releases VRAM	Model loaded, then unload()	model = None, gc.collect called
TC-AS-006	Pos	swap skips when config same	Same config twice	Second load skipped
TC-AS-007	Pos	truncates at max tokens	Long messages exceeding max_tokens	Dict with truncated content
TC-AS-008	Pos	_count_tokens from usage data	Chunk with usage.completion_tokens=42	Returns 42
TC-AS-009	Edge	_count_tokens fallback no usage	Empty chunks, content length 100	Returns 25 (len//4)
TC-AS-010	Pos	generate with GBNF grammar	Messages with grammar config	Returns dict

2.7 ClientConnection

File: app/modules/connection/__init__.py

Validate WebSocket messaging, heartbeat ping/pong, stale detection.

8 tests

ID	Type	Scenario	Input	Expected
TC-CC-001	Pos	send_status sends via WebSocket	Role=Planner, content=Planning..., phase=2	send_json called
TC-CC-002	Pos	send_result includes model names	planner=A, generator=B, judge=C	send_json with all 3 fields
TC-CC-003	Pos	send_halt_notification	No input	send_json with halt content
TC-CC-004	Pos	heartbeat starts background	Call start_heartbeat()	Task is not None
TC-CC-005	Pos	handle_pong resets counter	Counter at 1, call handle_pong()	_missed_pongs == 0
TC-CC-006	Pos	is_stale true after 2 missed	_missed_pongs=2	is_stale is True
TC-CC-007	Edge	is_stale false within limit	_missed_pongs=1	is_stale is False
TC-CC-008	Pos	send_status when fresh	is_stale=False, send_status()	send_json called

2.8 DatabaseManager

File: app/modules/context/__init__.py

Validate SQLite CRUD for session lifecycle, zombie cleanup, cascading deletes, retry.

12 tests

ID	Type	Scenario	Input	Expected
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TC-DB-001	Pos	create_session inserts row	session_id=s1, instruction=refactor, code=class A {}	user_instruction == refactor
TC-DB-002	Pos	complete_session updates	session_id=s1, exit=SUCCESS, code=class B {}	exit_status == SUCCESS
TC-DB-003	Pos	log_status creates log entry	session_id=s1, role=Planner, content=planning	1 log entry with backref
TC-DB-004	Pos	mark_as_halted sets status	session_id=s1	status == Halted
TC-DB-005	Pos	get_history ordered desc	Insert 3 sessions at different times	>=1 stubs, newest first
TC-DB-006	Pos	get_history_by_id with logs	Session + 2 log entries	Detail with 2 logs
TC-DB-007	Neg	get_history_by_id missing ID	ID nonexistent	None
TC-DB-008	Pos	rename_session updates title	session_id=s1, title=New Title	title == New Title
TC-DB-009	Pos	cleanup_zombie_sessions	Old unfinished session	Status updated to Zombie
TC-DB-010	Pos	cleanup_halted_sessions removes	Halted session, call cleanup	Session deleted
TC-DB-011	Pos	delete_history_by_id cascading	Session + 2 logs, delete	Session and logs gone
TC-DB-012	Edge	Tenacity retry on OperationalError	Transient DB failure	Retries and succeeds

2.9 MessageRouter

File: app/modules/connection/router.py

Validate WebSocket dispatch: pong, reconnect, refactor launch, halt, invalid payloads.

8 tests

ID	Type	Scenario	Input	Expected
TC-MR-001	Pos	dispatch routes pong	{type:pong}	client.handle_pong() called
TC-MR-002	Pos	dispatch routes reconnect	{type:reconnect, session_id:s1}	Reconnect handler called
TC-MR-003	Pos	dispatch launches single refactor	{type:single, code:class A {}, user_instruction:refactor}	Returns True
TC-MR-004	Pos	dispatch launches multi refactor	{type:multi, code:class A {}, user_instruction:refactor}	Returns True
TC-MR-005	Pos	dispatch routes halt	{type:halt}	agent.stop() called
TC-MR-006	Neg	dispatch rejects invalid request	{type:multi, code:}	Returns True (error sent)
TC-MR-007	Neg	dispatch rejects malformed JSON	Invalid payload	Returns False
TC-MR-008	Edge	dispatch ignores unknown type	{type:unknown}	Returns False

2.10 Orchestrator

File: app/modules/orchestrator/__init__.py

Validate orchestration state, feedback ring buffer, status notifications.

7 tests

ID	Type	Scenario	Input	Expected
TC-OR-001	Pos	OrchestrationState defaults	session_id=s1, base/working code, instruction	strategy_iter=1, exit_status=PROCESSING
TC-OR-002	Pos	add_feedback appends	0 feedback, add 1 entry	len == 1
TC-OR-003	Pos	add_feedback ring buffer capped	Add 4 entries	len == 3, most recent retained
TC-OR-004	Pos	extend_feedback batch cap	2 entries, extend 3 more	len == 3
TC-OR-005	Pos	Pipeline sends status notifications	Execute with mocked agent	send_status called >= 1
TC-OR-006	Pos	Status includes model names	Configured planner/generator/judge	Status sent with model info
TC-OR-007	Edge	Status skipped when stale	client.is_stale=True	send_status not called

2.11 Phase 2 — Strategy

File: app/modules/orchestrator/phases/phase2_strategy.py

Validate intent classification, analysis, synthesis, enrichment, deduplication, feedback translation.

12 tests

ID	Type	Scenario	Input	Expected
TC-P2-001	Pos	_classify returns IntentPacket	Mock returns FLATTEN_CONDITIONAL	state.intent_packet.specific_intent == FLATTEN_CONDITIONAL
TC-P2-002	Pos	_classify handles invalid intent	Mock returns unrecognized intent	State updated gracefully
TC-P2-003	Pos	_analyze stores analysis	Mock returns primary targets	state.architect_analysis populated
TC-P2-004	Edge	_analyze handles bad JSON	Mock returns malformed JSON	Graceful handling
TC-P2-005	Pos	_synthesize returns plan	Mock returns plan with mutations	state.active_plan has mutations
TC-P2-006	Pos	_synthesize retries invalid plan	First invalid, second valid	Retried, succeeds
TC-P2-007	Edge	_synthesize gives up max retries	All mocks return invalid	Pipeline aborts
TC-P2-008	Pos	_synthesize calls ASTMatcher	Plan with mutations	enrich_mutations called
TC-P2-008b	Pos	enrich dispatches all actions	ADD_FIELD, ADD_CONSTANT	Both processed

TC-P2-009	Pos	_deduplicate removes duplicates	3 mutations, 2 duplicates + 1 unique	len == 2
TC-P2-010	Pos	_deduplicate preserves unique	3 mutations, different targets	len == 3
TC-P2-011	Pos	_translate_feedback TIER_1	ValidationFinding with TIER_1_SYNTAX	Returns feedback hints

2.12 Phase 3 — Execution

File: app/modules/orchestrator/phases/phase3_execution.py

Validate generator output repair, multi-temperature CC selection, sequential mutation.

10 tests

ID	Type	Scenario	Input	Expected
TC-P3-001	Pos	repair strips throws Exception	Method with throws Exception added	Signature restored
TC-P3-002	Pos	repair strips null check	Generated null check guard	Null check removed
TC-P3-003	Pos	repair strips extra public	Spurious public on method	public not in result
TC-P3-004	Edge	repair leaves valid unchanged	Already matches original style	Result == original
TC-P3-005	Pos	run_single picks best CC	Mock returns CC 5, 3, 7 at 3 temps	CC=3 selected
TC-P3-006	Edge	run_single fallback all fail	All 3 outputs invalid	Falls back, continues
TC-P3-007	Pos	run_sequential applies one by one	2 mutations, both pass	working_code reflects both
TC-P3-008	Pos	sequential syntax healing	First mutation fails syntax	syntax_iter incremented
TC-P3-009	Edge	sequential boundary break	Mutation violates boundary	Sequential halted
TC-P3-010	Edge	sequential max heals exceeded	All mutations fail syntax	Pipeline handles

2.13 Phase 4 — Validation

File: app/modules/orchestrator/phases/phase4_validation.py

Validate validation chain (syntax, complexity, boundary, intent) and routing.

6 tests

ID	Type	Scenario	Input	Expected
TC-P4-001	Pos	run passes all checks	Valid code, CC decreased, no leaks	Validation passes
TC-P4-002	Neg	run routes to Phase 3 on syntax fail	Invalid Java code	syntax_iter incremented
TC-P4-004	Pos	complexity STRICT passes	CC decreases	Validation passes

TC-P4-005	Neg	complexity STRICT fails	CC same or increases	Validation fails
TC-P4-006	Pos	boundary passes	Non-target methods unchanged	Passes
TC-P4-007	Neg	boundary detects leak	Non-target also changed	Violation detected

2.14 Phase 5 — Adjudication

File: app/modules/orchestrator/phases/phase5_adjudication.py

Validate judge verdict routing, hallucination overrides, issue extraction.

5 tests

ID	Type	Scenario	Input	Expected
TC-P5-001	Pos	ACCEPT proceeds to Phase 6	{verdict:ACCEPT}	Phase 6 called
TC-P5-002	Pos	REVISE returns to Phase 2	{verdict:REVISE}	Routes to Phase 2, iter++
TC-P5-003	Edge	Identical code override	Working == base, no change	Override to REVISE
TC-P5-005	Edge	Missing response	Mock returns None	Assumes REVISE
TC-P5-006	Pos	Structured issues extracted	Verdict with issues	Issues parsed into state

2.15 Phase 6 — Finalization

File: app/modules/orchestrator/phases/phase6_finalization.py

Validate wrapper stripping, result transmission, LLM insights, DB persistence.

5 tests

ID	Type	Scenario	Input	Expected
TC-P6-001	Pos	Strips outer wrapper	Wrapped in fabricated class	Wrapper removed
TC-P6-002	Pos	Sends result with metrics	Valid state with working code	client.send_result() called
TC-P6-003	Pos	Generates insights	Mock returns insights	Insights generated
TC-P6-004	Edge	Insight failure graceful	Mock raises during insight gen	Continues, no crash
TC-P6-005	Pos	Persists to database	Valid state after pipeline	DB create/complete called

2.16 Formatters

File: app/utils/formatters.py

Validate conversion of JSON mutation plans to human-readable text.

6 tests

ID	Type	Scenario	Input	Expected
TC-FMT-001	Pos	Single MODIFY_METHOD	1 mutation on calculate	Text contains MODIFY_METHOD/calculate
TC-FMT-002	Pos	ADD_CONSTANT with value	MAX_SIZE=100 constant	Text contains MAX_SIZE
TC-FMT-003	Pos	Constant references in MODIFY	Modify references MAX_SIZE	Result not None
TC-FMT-004	Pos	Mutations ordered logically	RENAME, ADD_CONSTANT, MODIFY_METHOD	Logical order preserved
TC-FMT-005	Pos	Empty mutations message	{ast_mutations:[]}	Contains No mutations
TC-FMT-006	Edge	Empty mutations dict	{ast_mutations:[]}, minimal base	Result not None

2.17 Code Utils

File: app/utils/code_utils.py

Validate wrapper stripping, mutation ordering (rename-first, add-before-modify, stable sort).

7 tests

ID	Type	Scenario	Input	Expected
TC-CU-001	Pos	strip_outer_wrapper removes fabricated	public class Temp { void m() { } }	Temp not in result
TC-CU-002	Edge	strip_outer_wrapper preserves no wrapper	Already matches expected	Unchanged
TC-CU-003	Edge	strip_outer_wrapper class unchanged	Base is already a class	Unchanged
TC-CU-004	Pos	order_mutations rename first	[MODIFY_METHOD, RENAME_SYMBOL]	RENAME_SYMBOL first
TC-CU-005	Pos	order_mutations add before modify	[REMOVE_METHOD, ADD_DECLARATION, MODIFY_METHOD]	ADD before MODIFY/REMOVE
TC-CU-006a	Edge	order_mutations stable within group	2 MODIFY_METHOD, targets a,b	a before b preserved
TC-CU-006b	Edge	order_mutations stable sort 3	2 MODIFY_METHOD	Relative order maintained

3. Backend Integration Tests**3.1 API**

File: FastAPI application (app/main.py)

Validate REST API and WebSocket endpoints: health, history CRUD, real-time messaging.

8 tests

ID	Type	Scenario	Input	Expected
TC-IT-001	Pos	GET /health returns 200 with timestamp	—	Status 200, body has timestamp
TC-IT-002	Pos	GET /api/history returns empty list	No sessions	Status 200, body []
TC-IT-004	Neg	GET /api/history/:id 404 missing	ID invalid	Status 404
TC-IT-006	Neg	PATCH rejects empty title	Valid ID, title=	Status 400 or 422
TC-IT-008	Neg	DELETE 404 missing	ID invalid	Status 404
TC-IT-009	Pos	WebSocket /ws accepts connection	Connect via WS transport	Accepted
TC-IT-011	Pos	WS halt notification	Send halt message	Notification sent back
TC-IT-012	Neg	WS malformed JSON rejected	Invalid JSON	Rejected

3.2 Full Pipeline

File: tests/integration/test_full_mock_pipeline.py

Validate full 6-phase pipeline: SUCCESS, HALT, ABORT outcomes.

4 tests

ID	Type	Scenario	Input	Expected
TC-PL-SUCCESS	Pos	Pipeline completes 6 phases, persisted	BASE_CODE + INSTRUCTION, ACCEPT path	exit_status == SUCCESS, session Completed
TC-PL-RENAME	Pos	After pipeline: rename + delete	Same SUCCESS path	Rename succeeds, delete removes
TC-PL-HALT	Neg	User halt during generation	AbortingAgentService on 3rd call	exit_status == PROCESSING
TC-PL-ABORT	Edge	Strategy circuit breaker	BrokenCodeAgentService, invalid Java	exit_status == ABORT_STRATEGY, strategy_iter > 3

4. Frontend Unit Tests

4.1 lib/utls — cn()

File: src/lib/utls.ts

Validate Tailwind class merging, falsy filtering, conditional objects, twMerge.

6 tests

ID	Type	Scenario	Input	Expected
TC-	Pos	Merges multiple strings	cn(foo, bar)	foo bar

CN-001				
TC-CN-002	Pos	Filters falsy values	cn(foo, false, bar, undefined, null, baz)	foo bar baz
TC-CN-003	Pos	Handles conditional objects	cn(base, {active:true, disabled:false})	base active
TC-CN-004	Pos	Resolves Tailwind conflicts	cn(px-4, px-2)	px-2
TC-CN-005	Edge	No args returns empty	cn()	
TC-CN-006	Pos	Handles array arguments	cn([foo, bar], baz)	foo bar baz

4.2 lib/constants

File: src/lib/constants.ts

Validate constant shapes, role visuals, default fallback.

6 tests

ID	Type	Scenario	Input	Expected
TC-CNST-001	Pos	INITIAL_SOURCE empty string	—	
TC-CNST-002	Pos	EMPTY_ORCHESTRATION_RESULT shape	—	Has metrics, summary, diffHighlights
TC-CNST-003	Pos	ROLE_VISUALS has 6 roles	—	Exactly 6 keys
TC-CNST-004	Pos	Each role has step/icon/colorClass	—	All 18 fields present
TC-CNST-004b	Pos	Planner role values	—	{step:1, icon:Cpu, colorClass:text-[#56a8f5]}
TC-CNST-005	Pos	DEFAULT_ROLE_VISUALS fallback	—	{step:1, icon:Cpu, colorClass:text-jb-accent}

4.3 lib/parseStatusInfo

File: src/lib/parseStatusInfo.ts

Validate text parsing for phase numbers, iterations, retry, faults, decisions.

12 tests

ID	Type	Scenario	Input	Expected
TC-PS-001	Pos	parsePhaseNumber detects Ph1	Ph1: x	1
TC-PS-002	Pos	parsePhaseNumber detects Baseline	Baseline	1

TC-PS-003	Pos	parsePhaseNumber detects Strategy	Strategy	2
TC-PS-004	Pos	parseStrategyIteration extracts number	Strategy Iter 2	2
TC-PS-005	Pos	parseRetryInfo syntax heal	Syntax heal attempt 2/3	{current:2, max:3, type:syntax_heal}
TC-PS-006	Pos	parseRetryInfo sequential retry	Retry mutation 1/5	Object with current/max/type
TC-PS-007	Pos	parseValidationFaults count	Total Faults: 3	3
TC-PS-008	Pos	parseJudgeDecision ACCEPT	Decision: ACCEPT	ACCEPT
TC-PS-009	Pos	parseJudgeDecision REVISE	Verdict: REVISE	REVISE
TC-PS-010	Edge	parseIntentDetail unmatched	x	undefined
TC-PS-011	Edge	parseMutationPlan unmatched	x	undefined
TC-PS-014	Pos	parsePhaseAction extracts action	Ph2: Analyze	Analyze

4.4 lib/formatStatusContent

File: src/lib/formatStatusContent.ts

Validate extraction of JSON blocks, key-value tags, first-line summary.

8 tests

ID	Type	Scenario	Input	Expected
TC-FS-001	Pos	Extracts JSON blocks	Content with ``json...`` block	summary has values
TC-FS-002	Pos	Standalone JSON object	{"a":1}	Result defined
TC-FS-003	Pos	Key:Value tags from markdown	Category: CONTROL_FLOW\nMore	tags >= 1 entry
TC-FS-004	Pos	Markdown list input	- Intent: FLATTEN_CONDITIONAL	Result defined
TC-FS-005	Pos	First line summary	Start.\nMore	summary: Start.
TC-FS-006	Pos	Structured output properties	test	Has summary/tags/details
TC-FS-007	Edge	Empty input		summary:
TC-FS-008	Edge	Plain text input	plain	summary: plain

4.5 lib/javaFormatter

File: src/lib/utils/javaFormatter.ts

Validate Java formatting: operator spacing, brace merging, generics, comments.

9 tests

ID	Type	Scenario	Input	Expected
TC-JF-001	Pos	Formats basic class	class A { void m() { } }	Truthy output
TC-JF-002	Pos	Preserves string literals	Code with select * from table	hello preserved
TC-JF-003	Pos	Preserves comments	Code with // comment	Comment preserved
TC-JF-004	Pos	Handles generic types	List<Map<String, Integer>>	Truthy output
TC-JF-005	Pos	Normalizes operator spacing	a=b+c	No a=b+c in output
TC-JF-006	Pos	Merges else/catch/finally	} \else {	} else { merged
TC-JF-007	Edge	Strips consecutive blank lines	3 blank lines	Single blank line
TC-JF-009	Edge	Handles empty string		
TC-JF-010	Pos	Long method mixed constructs	Conditionals, loops, try/catch	Truthy output

4.6 lib/indentation

File: src/lib/indentation.ts

Validate smart Enter, Tab/Shift-Tab, closing brace alignment.

7 tests

ID	Type	Scenario	Input	Expected
TC-IN-001	Pos	handleEnterKey adds indent after line ending {	if(x) { at end	New line with indent
TC-IN-002	Pos	handleEnterKey auto-closes brace	main() { at end	Inserts \n \n}
TC-IN-003	Edge	handleEnterKey no double close	Line already has }	Only indents
TC-IN-004	Pos	handleTabKey indents	foo(); at start	foo();
TC-IN-006	Pos	handleShiftTab outdents	foo();	foo();
TC-IN-007	Pos	handleClosingBrace aligns to opening	if(x) {\n } (cursor after)	Adjusted to match
TC-IN-008	Edge	handleClosingBrace null when correct	} already aligned	null

4.7 lib/buildMetrics

File: src/lib/utis/buildMetrics.ts

Validate CC metrics (direction up/down/neutral), inference time, GPU metrics.

6 tests

ID	Type	Scenario	Input	Expected
TC-BM-001	Pos	Complexity decreased	orig=10, refactored=5	direction: down, before:10, after:5
TC-BM-002	Pos	Complexity increased	orig=5, refactored=10	direction: up
TC-BM-003	Pos	Complexity unchanged	orig=5, refactored=5	direction: neutral
TC-BM-004	Pos	Inference time metric	performance inference_time=12.5	after: 12.5s
TC-BM-005	Pos	GPU metrics with performance	performance with avg_gpu_utilization	>=3 total metrics
TC-BM-006	Edge	No GPU when null performance	performance=null	Only 1 metric

4.8 lib/schemas/websocket (Zod)

File: src/lib/schemas/websocket.ts

Validate Zod schema validation for all WS message types, discriminated union.

8 tests

ID	Type	Scenario	Input	Expected
TC-ZD-001	Pos	StatusMessageSchema valid	{type:status, role:Planner, content:...}	Parses, type == status
TC-ZD-002	Neg	StatusMessageSchema rejects missing role	{type:status, content:...}	Throws
TC-ZD-003	Neg	StatusMessageSchema rejects invalid type	{type:invalid, role:Planner, content:...}	Throws
TC-ZD-004	Pos	ResultMessageSchema valid	{type:result, code:class A {}, exit_status:SUCCESS}	Parses, type == result
TC-ZD-005	Neg	ResultMessageSchema invalid exit_status	{type:result, code:..., exit_status:INVALID}	Throws
TC-ZD-006	Pos	ErrorMessageSchema valid	{type:error, message:Something went wrong}	Parses, type == error
TC-ZD-007	Pos	ServerMessageSchema discriminates	Status/result/error objects	Each matches correct schema
TC-ZD-008	Neg	ServerMessageSchema rejects unknown	{type:unknown}	Throws

4.9 store/useChatStore

File: src/store/useChatStore.ts

Validate Zustand store: session CRUD, fetch/rename/delete, migration, draft.

15 tests

ID	Type	Scenario	Input	Expected
TC-ST-001	Pos	createSession creates with defaults	createSession(test-1)	title: New Session, empty entries
TC-ST-002	Pos	createSession with initial prompt	createSession(test-1, refactor)	Session created, ID defined
TC-ST-003	Pos	updateSession partial merge	Create, update title	Title changed, others unchanged
TC-ST-004	Pos	updateSession updater accumulates	2 pushes to terminalEntries	len == 2
TC-ST-005	Pos	deleteSession calls API + removes	Session exists, mock 200	Session removed
TC-ST-006	Edge	deleteSession rollback on failure	Mock 500	Session restored
TC-ST-007	Pos	renameSession optimistic update	Mock PATCH 200	Title updated
TC-ST-008	Edge	renameSession rollback on fail	Mock PATCH 500	Title reverted
TC-ST-009	Pos	migrateSessionId preserves data	old-id to new-id	Old removed, new has same data
TC-ST-010	Pos	fetchHistory populates	Mock GET 2 stubs	2 sessions
TC-ST-011	Neg	fetchHistory error flag	Mock GET error	historyLoadError == true
TC-ST-012	Pos	fetchSessionDetails maps logs	Valid session ID	Session populated
TC-ST-013	Pos	resetDraftSession clears	Draft has values, reset	Back to defaults
TC-ST-014	Pos	setOrchestratorStatus updates	connected	orchestratorStatus == connected
TC-ST-014b	Pos	hasInitialLoaded tracks state	After fetchHistory resolves	false then true

4.10 hooks/useOrchestrationSocket

File: src/hooks/useOrchestrationSocket.tsx

Validate WebSocket lifecycle, method availability.

3 tests

ID	Type	Scenario	Input	Expected
TC-WS-001	Pos	Returns connection status	Render hook	connectionStatus defined
TC-WS-001	Pos	Exposes required methods	Render hook	Has connect, sendRefactorRequest,

b				sendHaltRequest
TC-WS-002	Pos	disconnect changes status	Call disconnect()	Status updated

4.11 components/ErrorHandler

File: src/components/layout/ErrorHandler.tsx

Validate React error boundary: rendering, catching, fallback, reset.

4 tests

ID	Type	Scenario	Input	Expected
TC-EB-001	Pos	Renders children when no error	<ErrorHandler> ok </ErrorHandler>	Shows ok
TC-EB-002	Pos	Catches error, shows fallback	Child throws new Error(crash)	Shows Something went wrong
TC-EB-003	Pos	Shows custom fallback	fallback=<CustomFallback /> + crashing child	Shows CustomFallback
TC-EB-004	Edge	Resets when children replaced	Catch error, re-mount non-crashing child	Shows new children, error cleared

5. Frontend Integration Tests

5.1 Terminal

File: src/components/features/terminal/Terminal.tsx

Validate rendering of terminal entry types, collapsible header.

3 tests

ID	Type	Scenario	Input	Expected
TC-FI-001	Pos	Renders boot header	Initial render	Welcome to Horizon AI shown
TC-FI-002	Pos	Renders terminal entry types	Store has command/log/system/error entries	Entry content rendered
TC-FI-003	Pos	Collapsible header visible	Initial render	Terminal header rendered

5.2 ChatWorkspace

File: src/components/features/workspace/ChatWorkspace.tsx

Validate panel layout rendering with valid/null session ID.

2 tests

ID	Type	Scenario	Input	Expected
TC-FI-004a	Pos	Renders with valid session ID	sessionId=test-session	Body truthy
TC-FI-004b	Pos	Renders with null session ID	sessionId=null	Body truthy